

Grade 8
Unit 4
Lesson 9 Practice

Name _____
Date _____
Period _____

For each expression in questions 1-8, generate an equivalent expression with positive exponents by applying one or more laws of exponents.

1. 4^{-3}

$$\frac{1}{4^3}$$

2. $(d)^{-9}$

$$\frac{1}{d^9}$$

3. g^{-4}

$$\frac{1}{g^4}$$

4. $\left(\frac{1}{a}\right)^{-6}$

$$a^6$$

5. $\left(\frac{6}{t}\right)^{-7}$

$$\frac{t^7}{6^7}$$

6. $\frac{7k^{-8}}{k^2}$

$$\frac{7}{k^{10}}$$

7. $\frac{9x^9}{x^{15}}$

$$\frac{9}{x^6}$$

8. $-5y^{-5} \cdot -8y^{-1}$

$$\frac{40}{y^6}$$

Find the value of r that makes the equation true for questions 9–10.

9. $\frac{x^{13}}{x^r} = x^5$ $r = 8$

10. $\frac{x^3}{x^r} = x^9$ $r = -6$